

BHUVANESH A

AI & Machine Learning · Software Development (Learner) · Research

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PROFILE SUMMARY

AI & ML student with strong technical foundations in programming, machine learning, and full-stack development, complemented by clear communication skills and the ability to present technical ideas effectively through projects and collaboration.

EDUCATION

Coimbatore Institute of Technology — MSc Artificial Intelligence & Machine Learning	2024 – 2029
GPA: 8.27 (Sem 1) 8.50 (Sem 2) 7.86 (Sem 3)	
The Frontline Academy — Higher Secondary Certificate (Class XII)	2023 – 2024
Aggregate: 87%	

TECHNICAL SKILLS

Programming	Python, Java, C, SQL
ML / AI	Scikit-learn, LightGBM, XGBoost, NLP, PCA / LDA, Regression, Classification
Data & Viz	Pandas, NumPy, Power BI, Matplotlib, Seaborn
Web / Full-Stack	HTML, CSS, JavaScript, Flask, SQLite
Tools & Platforms	Figma, Git, GitHub, Ollama, Flutter (basic)
Soft Skills	Technical Communication, Research Writing, Collaboration, Problem Solving

PROJECTS

Candida Auris Infection Risk Intelligence System

Stack: Python · LightGBM · Flutter

- Trained a LightGBM classifier on 10+ clinical features to predict *Candida auris* infection risk, achieving 91% accuracy on held-out test data.
- Reduced false-negative rate by 18% over baseline logistic regression through iterative feature engineering on an imbalanced medical dataset.
- Deployed predictions to a cross-platform Flutter dashboard, cutting manual risk-assessment time from ~30 min to under 2 min per patient.

Network Intrusion Detection System with ML Dashboard

Stack: Python · Scikit-learn · Random Forest · Data Visualization

- Built a 5-class supervised ML pipeline on 80,000+ network traffic records, achieving 95%+ classification accuracy across benign and 4 attack categories.
- Automated dashboard generation displaying threat patterns, top-10 feature importances, and confusion matrix — reducing analyst triage time by ~40%.
- Evaluated multiple ML algorithms and selected the best-performing model based on classification accuracy.

Fitness Guide Chatbot

Stack: Python · Ollama (LLaMA 3) · Flask · HTML · CSS · JavaScript

- Built a locally-hosted LLM chatbot using Ollama (LLaMA 3) that answers personalised fitness queries with sub-2s response latency.
- Ran the model locally with no API cost, enabling offline usage and multi-turn conversation support.

URL Shortener Service

Stack: Flask · SQLite · HTML/CSS

- Engineered a URL shortening service handling 500+ test redirects with collision-resistant 6-character hash generation.
- Reduced average redirect lookup time to under 5ms using indexed SQLite queries and in-memory caching.

Gym Membership Management System

Stack: Java · SQL

- Designed a normalised 6-table relational schema supporting 3 membership tiers and 200+ member records.
- Implemented full CRUD operations with basic membership expiry tracking.

ACHIEVEMENTS & COMPETITIONS

- **Smart Ideation Ideathon (SIH Pre-event) — 3rd Place** — competed in the ideation round and secured 3rd place.
- **Kaggle ML Competition — Predicting BPM of Songs:** Participated in a machine learning competition involving data preprocessing, dimensionality reduction, and regression modelling using Python.
- **LeetCode:** Solved 50+ problems (arrays, linked lists, stacks, trees) with an average runtime faster than 70% of submissions.
- **Smart India Hackathon (SIH) 2025** — Participated at national level (Certificate of Participation).

CERTIFICATIONS & TRAINING

- Deloitte Data Analytics Job Simulation — *Deloitte (Forage)*
- Natural Language Processing Workshop — *Dept. of AI & Data Science, CIT*
- Generative AI & Agentic AI: Building Autonomous Systems (One-Week Virtual FDP) — *Coimbatore Institute of Technology*
- Statistela: Data Analytics Competition — *B.A.S.H 8.0 (Certificate of Participation)*